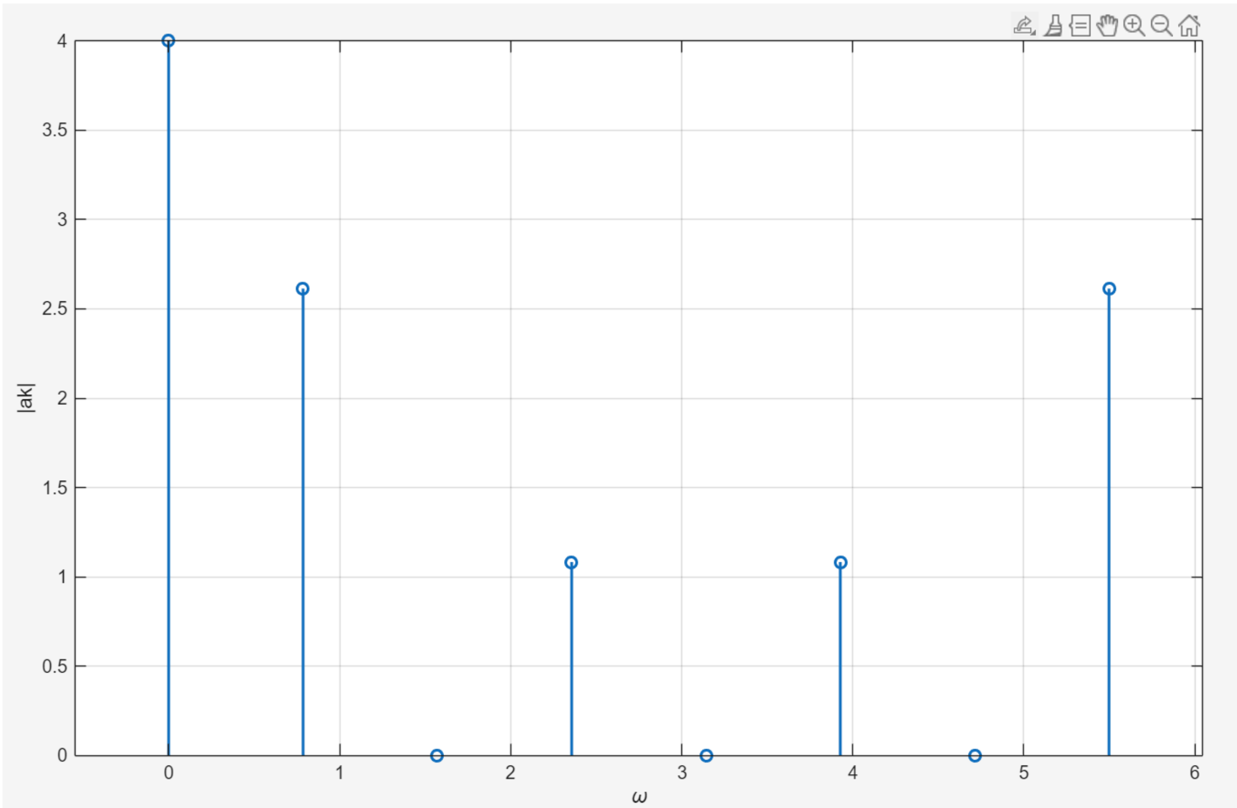
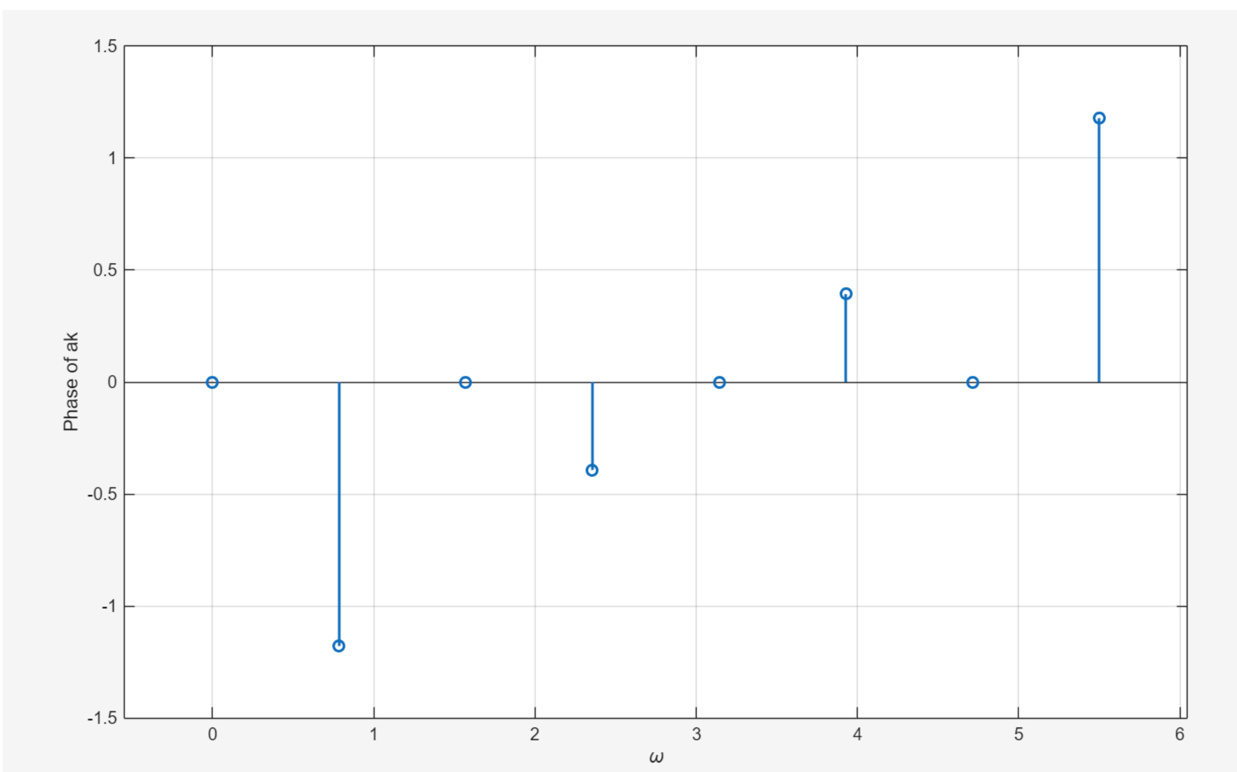
# EE110B Lab#5

Nathan Taing

## Plotting DTFS Coefficients

1)



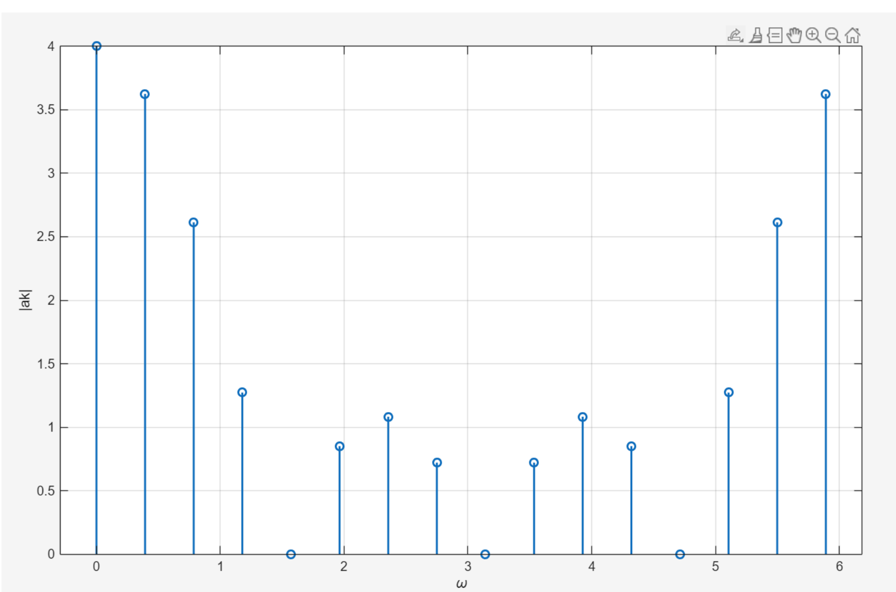



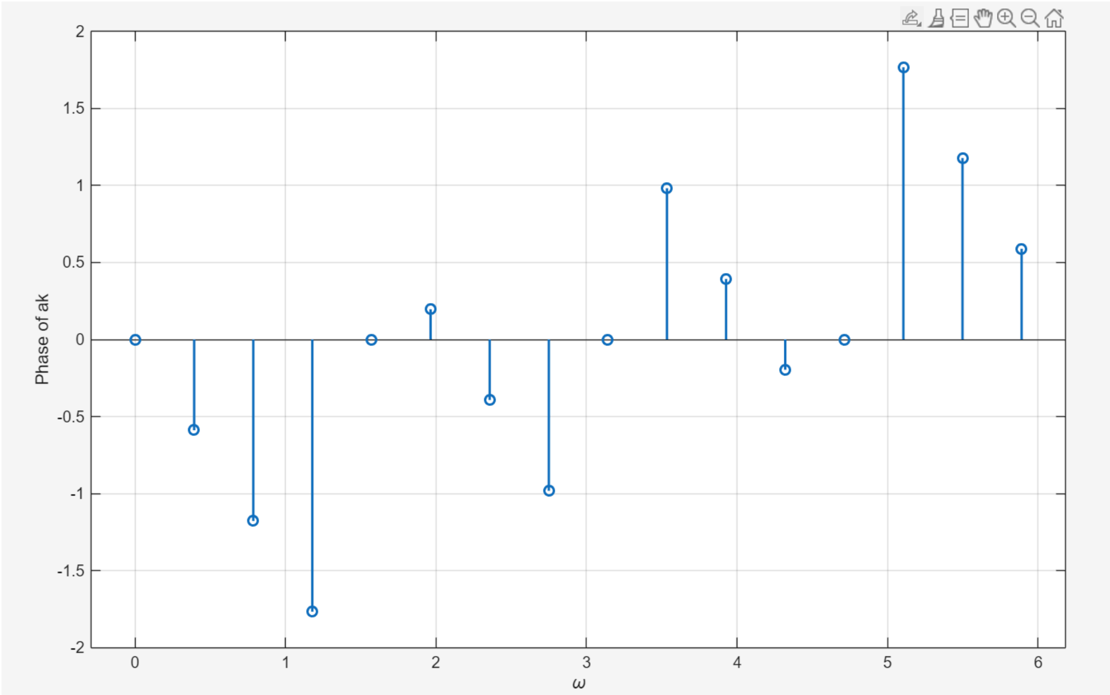
Code:

```
N = 8;
x_arr = [1 1 1 1 0 0 0 0];
x_dtfs = fft(x_arr);
dtfs_freqs = 2*pi*(0:(N-1)) / N;
figure;
stem(dtfs_freqs, abs(x_dtfs), 'LineWidth', 1.5);
xlabel('\omega');
ylabel('|ak|');
grid on;
figure;
stem(dtfs_freqs, angle(x_dtfs), 'LineWidth', 1.5);
xlabel('\omega');
ylabel('Phase of ak');
grid on;
```

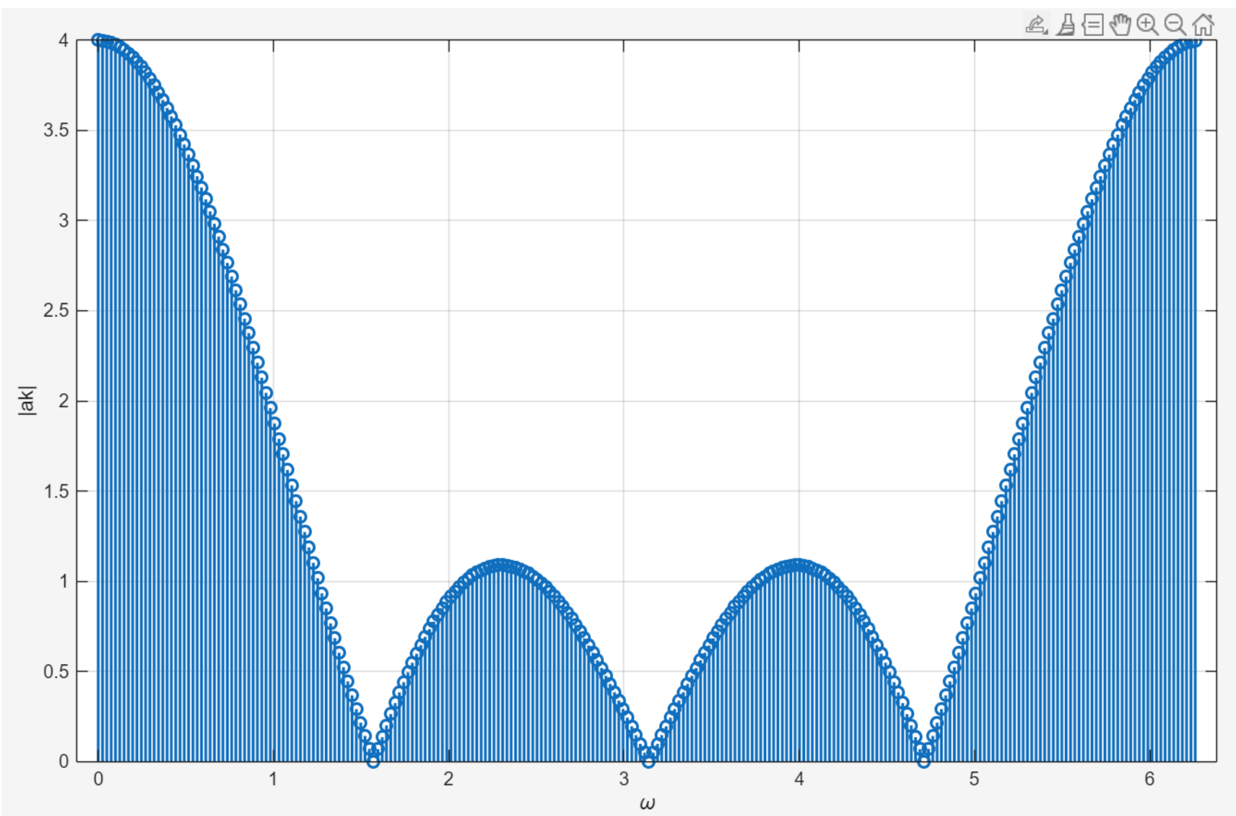
These figures above plot both the magnitude and the phase of the DTFS of our original x_arr signal. As we can see in the magnitude above, our signal contains a strong DC component. This is because the original x_arr is the value 1 for half of the time, giving the signal a positive average value. The phase graph also demonstrates a symmetric pattern. This is because of the real-valued nature of the signal. When you put the magnitude and the phase together, you attain coefficients that fully describe the periodic signal in the frequency domain.

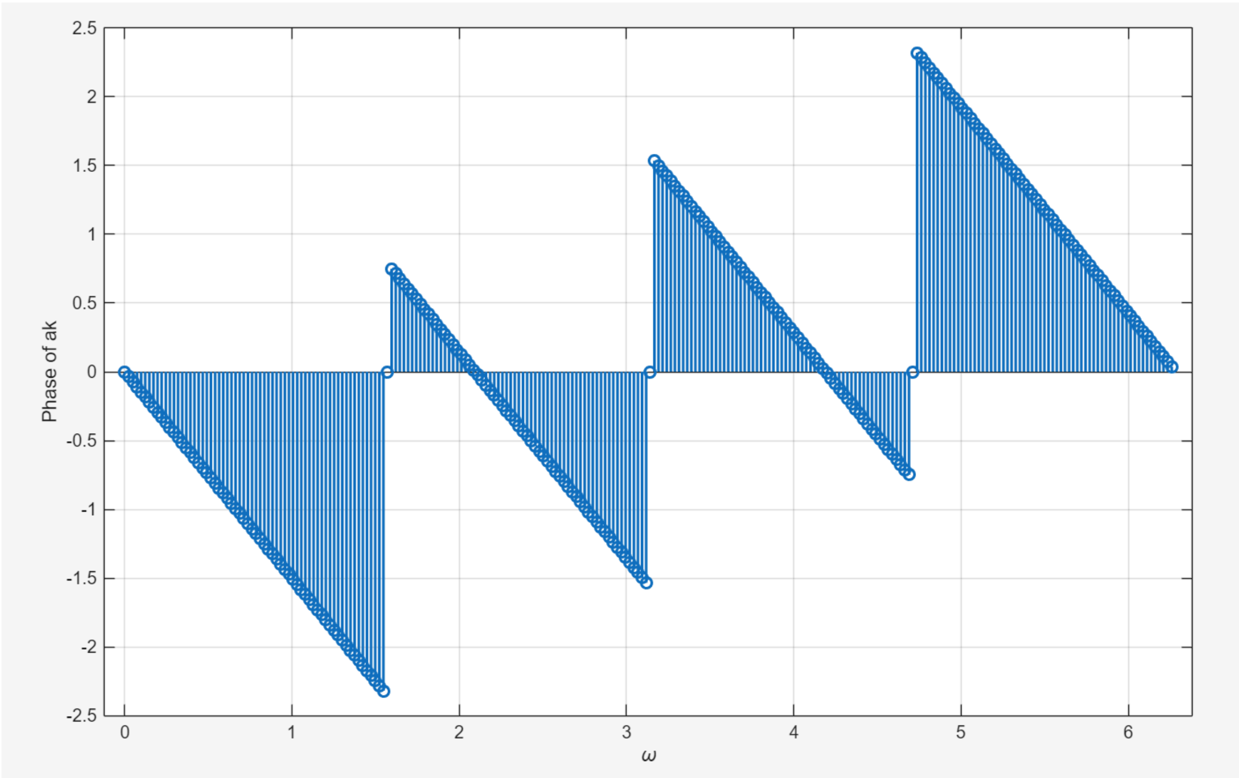
2) Same code, but we changed the value of N and we extended x_arr to be the same length.
N = 8






N = 256






Increasing N allows us to gain a better understanding of the magnitude and the phase of the signal. The spacing decreases between each coefficient, allowing us to view the signal better. Doubling the N allowed us to get a clearer view of the shape of the x_arr in the frequency domain. We can see the influence the phase and magnitude have together to create the shape of the signal. However, when we boost N up to 256, we get the full picture of the signal in the frequency domain. Each of the DTFS coefficients are symmetric, meaning the original signal is real valued. When reconstructing the original signal, the conjugate frequency components cancel out, leaving the reconstructed signal.